Plotting coordinates generated from a rule

Task A: Finding a rule from a set of coordinates

1) For each set of coordinates, fill in the gaps to describe the relationship in words and using algebra:

a) (0,1) (2,3) (9,10)

The x coordinate is equal to the y coordinate *plus 1*.

$y = x + 1$

b) (2,6) (-4,0) (4,8)

The x coordinate is equal to the y coordinate plus 4.

$y = x + 4$

c) (10,5) (4,-1) (0,-5)

The x coordinate is equal to the y coordinate *minus 5*.

$y = x - 5$

d) (-7,-2) (-3,2) (2.2,7.2)

The x coordinate is *equal to the y coordinate plus 5*.

$y = x + 5$

e) (2,4) (3,6) (8,16)

The x coordinate is *double the y coordinate*.

$y = 2x$

f) (2,20) (-4,-40) (3,30)

The x coordinate is *ten times the y coordinate*.

$y = 10x$

g) (0,0) (1,5) (-1,-5)

The x coordinate is *the y coordinate multiplied by 5*.

$y = 5x$

h) (10,1) (6,0.6) (30,3)

The x coordinate is *the y coordinate divided by 10*.

$y = \frac{1}{10}x$ or $y = \frac{x}{10}$

Task B: Generating coordinates from a rule

1) Generate 3 coordinates that follow each rule. Some have the x or y coordinate already filled in.

There are many possible answers for the blank coordinates. Examples:

a) (3 , 23) (,) (,) (0,20) (1,21) (10,30) (-20,0)

$y = x + 20$

b) (0 , -9) (18 , 9) (,) (1,-8) (10,1) (-2,-11) (100,91)

$y = x - 9$

c) (0 , 0) (3 ,300) (,) (1,100) (0.1,10) (10,1000) (-3,-300)

$y = 100x$

d) (0 , 0) (,) (,) (1,4) (2,8) (25,100) (-3,-12)

$y = 4x$

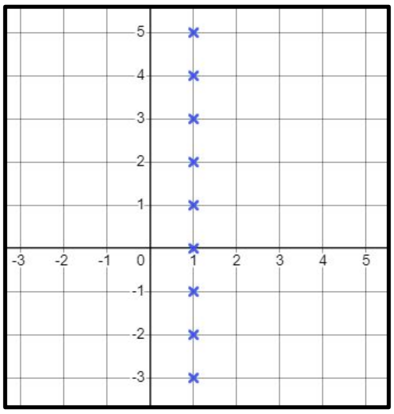
2) Fill in the table with a different coordinate for the equation that satisfies each condition. The first one has been done for you.

		The y coordinate is positive	The y coordinate is negative	The x coordinate is a non-integer
a)	$y = x - 5$	(15,10)	(-5,-10)	(7.5, 2.5)
b)	$y = x + 10$	E.g (-4, 6)	E.g (-20, -10)	E.g (0.2, 10.2)
c)	$y = 5x$	E.g (2, 10)	E.g (-10, -50)	E.g (0.1, 0.5)
d)	$y = -3x$	E.g (-3, 9)	E.g (2, -6)	E.g (0.5, -1.5)
e)	$y = x - 0.02$	E.g (6,5.98)	E.g (-5, -5.02)	E.g (0.52, 0.5)
f)	$y = \frac{x}{3}$	E.g (9, 3)	E.g (-12, -4)	E.g (3.9, 1.3)

Task C: Plotting coordinates that follow a rule

1) Plot all the integer points that satisfy each rule.

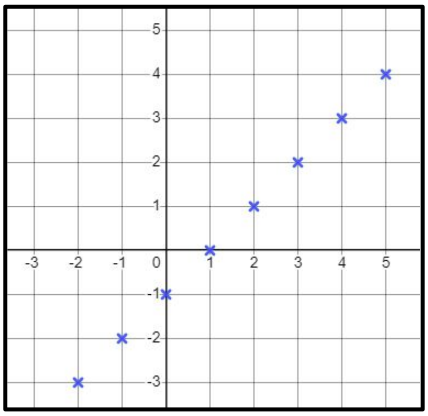
a) The x value is 1



Describe the pattern

A vertical line

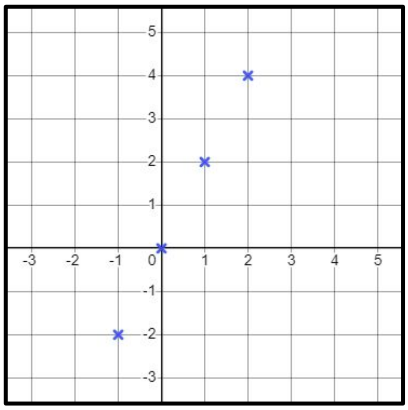
b) The y value is one less than the x value



Describe the pattern

A diagonal line

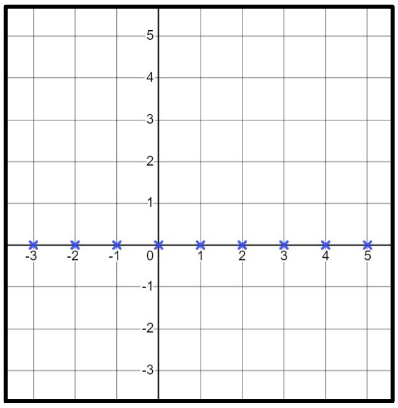
c) The y value is double the x value



Describe the pattern

A diagonal line

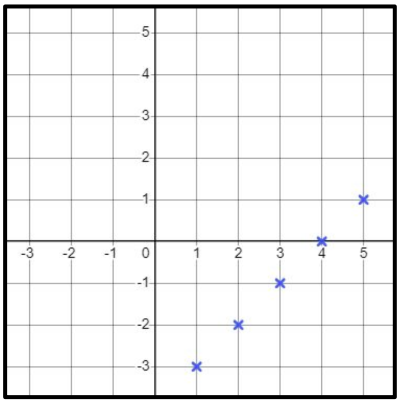
d) $y = 0$



Describe the pattern

A horizontal line- the x *axis*

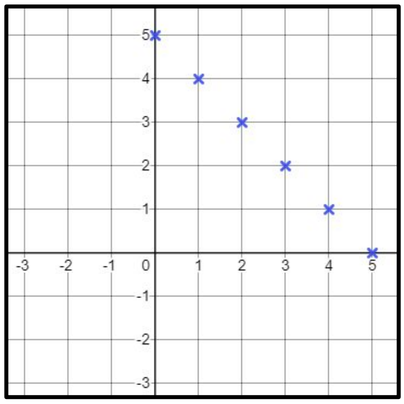
e) $y = x - 4$



Describe the pattern

A diagonal line

f) $x + y = 5$



Describe the pattern

A diagonal line

2) The grid shows seven points labelled A, B, C, D, E, F and G. Complete the table to show which points have coordinates that match the rules below. The first column has been completed for you

Rule	A	B	C	D	E	F	G
$x = 2$	yes	*yes*	*no*	*no*	*no*	*no*	*no*
$y = -2$	no	*yes*	*no*	*yes*	*no*	*no*	*yes*
$x = y$	yes	*no*	*no*	*yes*	*yes*	*yes*	*no*
$y = -x$	no	*yes*	*yes*	*no*	*yes*	*no*	*no*